Diffusion Models in Finance: A Survey

Anonymous Author(s)
Anonymous Institution(s)

ABSTRACT

Diffusion and score-based generative models have, within a few years, spread across many major quantitative-finance tasks: synthetic data generation, limit- order-book (LOB) simulation, risk and scenario generation, derivative pricing and hedging, and portfolio construction. Their appeal is structural—stable likelihood-based training, strong mode coverage, native conditioning, and a stochastic-differential-equation (SDE) formulation that sits naturally beside the Itô calculus and stochastic control finance already uses. We argue that progress is now bottlenecked less by architecture than by finance-specific evaluation, constraints, and deployment validation. We present, to our knowledge, the first survey focused specifically on diffusion-family generative models across financial tasks. We organize a fast-growing and fragmented literature along two axes—methodological family and financial task—consolidate how diffusion's conditioning and SDE structure are exploited for finance-specific goals (risk-neutral generation, stylized-fact fidelity, controllable stress, microstructure realism), and contribute a critical review of evaluation together with a concrete reporting protocol, since heterogeneous metrics, leakage, and in-distribution testing currently limit comparability. We close with a research agenda: tail fidelity, non-stationarity guarantees, hard economic constraints, efficiency for real-time use, privacy and governance, and standardized benchmarks.

KEYWORDS

diffusion models, score-based generative models, synthetic data, market simulation, limit order book, risk management, derivative pricing, portfolio optimization, stress testing

1 INTRODUCTION

Generative modeling is now central to quantitative finance, for structural rather than fashionable reasons. Financial data are *scarce* in the regimes that matter most—crises, regime transitions, and rare events are by definition under-represented in any historical sample. They are *sensitive*, since client-level and proprietary records cannot be shared freely. And they are *non-stationary*, drifting with policy, liquidity, and market structure. Synthetic data that faithfully reproduces the statistical signature of markets relaxes all three constraints at once: it augments thin datasets, substitutes for privacy-restricted records, and supplies controllable scenarios for backtesting and stress testing [33, 61].

For roughly a decade this need was met by GANs and VAEs: Quant GANs [79], signature-based Sig-Wasserstein GANs [54], and structured-noise/MMD models [53] reproduce volatility clustering and fat tails. Yet GANs remain hard to train, prone to mode collapse, and awkward to condition, while VAEs oversmooth the heavy tails that dominate financial risk.

Why diffusion, why now. Denoising diffusion probabilistic models (DDPMs) [31] and score-based models [68], unified under a stochastic differential equation (SDE) view [69], offer a different trade-off: stable likelihood-based training, strong mode coverage, and native, flexible *conditioning* through guidance [20, 32]. They also sit beside the objects quantitative finance already uses: a diffusion model *is* an SDE with a learned drift, sampling is reverse-time simulation, and training has a control-theoretic Hamilton–Jacobi–Bellman (HJB) reading [9]. For a field built on Itô calculus, Monte Carlo, and stochastic control, this is a comfortable fit—and the reason diffusion methods have, since roughly 2023, spread across many major quantitative-finance tasks.

Contributions. Our central contribution is normative: (i) a critical review of evaluation and a concrete *reporting protocol* (§10), because the field's binding constraint is measuring progress, not producing methods. Around it we offer (ii) a two-axis taxonomy (method × financial task, Figure 1) and coverage map (Figure 2); (iii) a consolidated account of how diffusion's conditioning and SDE structure serve finance-specific goals—risk-neutral generation, stylized-fact fidelity, controllable stress, and microstructure realism; and (iv) a research agenda. The *delta* over generic time-series diffusion surveys [51, 81] and broad GenAI/tabular synthetic-data surveys [13, 46] is the finance-specific structure those reviews do not treat: the $\mathbb{P} \to \mathbb{Q}$ measure change and no-arbitrage, microstructure *responsiveness*, tails as a capital quantity, the simultaneous battery of stylized facts, and privacy/governance. To our knowledge this is the first survey focused specifically on diffusion-family generative models across financial tasks, rather than on time-series diffusion, tabular synthesis, GenAI-in-finance, or portfolio/forecasting use alone. Section 2 reviews fundamentals; Sections 3–7 survey applications; Section 8 covers emerging and cross-domain uses; Section 9 synthesizes cross-cutting themes; Section 10 treats evaluation; and Section 11 discusses open problems.

Scope. We focus on diffusion and score-based models—DDPMs, continuous score-SDEs, and their conditional, latent, fast-sampling, and structure-informed variants—applied to financial problems. We also include the *flow-matching*, *rectified-flow*, and *stochastic-interpolant* generalization (§2), of which diffusion is the Gaussian special case, but treat classical normalizing flows as a related baseline. We include the most relevant non-diffusion baselines (notably GANs and VAEs) for context but do not survey them exhaustively, and we treat cross-domain backbones only insofar as they are adapted to finance. Within each financial task we organize work by method. We prioritize peer-reviewed and widely cited results, while noting that much of this literature is recent and preprint-stage, so quantitative claims should be read as indicative.

Methodology. We paired method terms (*diffusion*, *score-based*, *flow matching*, *rectified flow*, *stochastic interpolant*) with finance terms (*market*, *limit order book*, *volatility*, *portfolio*, *risk*, *option pricing*, *tabular financial data*) across arXiv, SSRN, ACM, IEEE, and Google Scholar through 2026 Q1, then snowballed citations. From 103 recorded candidate/seed records, we retained the 88 cited works

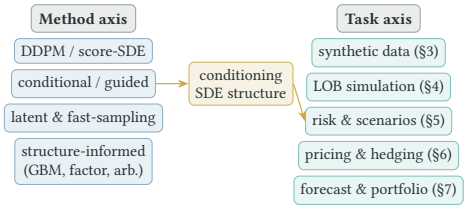


Figure 1: Two-axis taxonomy. Each surveyed work pairs a methodological family with one or more financial tasks; conditioning and SDE structure are the threads that recur across both axes.

applying diffusion-family generators or generative decision models to financial tasks, excluded three false positives, and replaced eleven early candidates after verification. We coded primary task, method family, modality, conditioning, evaluation type, publication status, and verifiable code/data availability; Figure 2 aggregates these primary codes and Table 1 lists representative entries. The corpus is a preprint-heavy snapshot, biased toward published, well-cited, thesis-confirming work; negative results are likely under-represented (§10).

2 BACKGROUND

This section gives the minimum machinery to read the survey and, more importantly, draws the bridge to financial mathematics.

Forward and reverse processes. A diffusion model defines a *forward* process that corrupts data $x_0 \sim p_{\text{data}}$ into noise and learns a *reverse* process that reconstructs data from noise. In DDPM [31], the forward chain adds Gaussian noise,

$$q(x_t \mid x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}\, x_{t-1}, \beta_t I),$$
$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon.$$

A network $\epsilon_\theta(x_t, t)$ is trained with the denoising objective $\mathbb{E}\|\epsilon - \epsilon_\theta(x_t, t)\|^2$, an idea rooted in non-equilibrium thermodynamics [65]. Score-based models [68] instead learn the score $\nabla_x \log p_t(x)$ by denoising score matching across noise scales; predicting noise and predicting the score are equivalent up to scaling. The simple objective is a reweighting of the variational bound on $\log p_\theta(x_0)$, so training is likelihood-based and stable—a sharp contrast with the adversarial min–max of GANs, and a practical reason diffusion has displaced them in finance, where training instability and mode collapse directly translate into missing market regimes. Conditioning on covariates c (a regime label, a volatility target, a history) is handled by learning $\epsilon_\theta(x_t, t, c)$ and, at sampling time, combining conditional and unconditional predictions via classifier-free guidance [32],

$$\tilde{\epsilon} = \epsilon_\theta(x_t, t, \varnothing) + w\,[\epsilon_\theta(x_t, t, c) - \epsilon_\theta(x_t, t, \varnothing)].$$

The guidance weight w trades fidelity to c against diversity—the knob that makes "generate a crash of intensity c" operational.

The SDE unification. Song et al. [69] unify both under an Itô SDE $dx = f(x,t)\,dt + g(t)\,dW_t$ whose time reversal, $dx = [f(x,t) - g(t)^2 \nabla_x \log p_t(x)]\,dt + g(t)\,d\bar{W}_t$, is itself driven by the score, with a deterministic probability-flow ODE of identical marginals. *This is the heart of the finance connection*: asset-price models are SDEs, so a diffusion generator is a learned, data-driven SDE, and sampling is the reverse-time simulation quants already perform—now with a neural drift. The correspondence is operational, not merely aesthetic: it tells us where financial structure can be injected (the drift f, the diffusion g, the noise schedule, or the prior), how to reason about a change of measure (a drift shift, hence a score shift; §6), and why control-theoretic tools apply—the training objective satisfies an HJB equation on the evolving log-density [9], connecting generation to the optimal-investment and hedging problems of continuous-time finance—though, to date, this is mainly a unifying lens rather than a tool the surveyed methods exploit concretely (§11). The probability-flow ODE additionally yields exact likelihoods, useful for anomaly scoring and model comparison.

Discrete vs. continuous. The two are views of one object, but the choice matters in finance. The discrete DDPM suits fixed-length tabular records and short return windows [42, 61]; the continuous SDE view is natural when the target is itself a process—asset paths, yield curves, order flow—since it exposes the drift and diffusion terms where economic structure (mean reversion, jumps, the risk-neutral shift) is imposed [24, 40, 74]. Few-step distillations and latent variants apply to either. In practice models mix these freely: a latent score model with a transformer backbone, classifier-free conditioning, and a finance-specific prior is a common composite.

From recipe to practice. Financial diffusion models repeatedly choose a backbone (Transformer, U-Net, state-space), representation (series, latent, image, byte stream), conditioning route (concatenation, cross-attention, guidance), noise schedule, and structural prior. These choices transfer once the modality is matched, but careless transfer imports assumptions—Gaussianity, stationarity, exchangeability—that financial data violate.

Sampling and acceleration. Ancestral sampling needs hundreds to thousands of network evaluations, prohibitive for high-frequency or large-scale risk use. Refinements [55]—DDIM [66], EDM [39], consistency models [67], flow matching [49]—push toward few-step generation, a binding constraint we revisit below.

Flows and interpolants: the broader family. Flow matching is the entry point to a *generalization* of diffusion we treat as in scope. Rectified flow [52] learns a neural ODE transporting noise to data along near-straight paths, and stochastic interpolants [3] unify the picture: a tunable interpolation between any two densities yields both a probability-flow ODE and a family of SDEs, with classical diffusion the Gaussian special case. This matters in finance because arbitrary base distributions and schedules let priors be built into the *path*, and the ODE form supports few-step sampling where latency binds. We therefore include flow/rectified-flow/interpolant methods alongside DDPM and score-SDE, treating *classical* normalizing flows as a related baseline.

Conditioning and backbones. Finance rarely wants unconditional samples; it wants samples *given* a regime, a volatility target, or a history. Classifier [20] and classifier-free [32] guidance steer generation; latent diffusion [59] improves efficiency. Cross-domain backbones recur as finance models and baselines: TimeGrad [58], CSDI [73], SSSD [4], Diffusion-TS [82], TSDiff [41] for sequences; TabDDPM [42], TabSyn [84] for tables; DiT [57] for transformer backbones; diffusion planning [2, 37] for decisions; and flow-family backbones, FlowTS [34], TimeFlow [29], interpolant forecasting [15, 16], and RecTable [21].

Why finance is hard. Generic successes do not transfer automatically: returns are heavy-tailed, volatility-clustered, and non-stationary, the signal-to-noise ratio is low, LOB data are discrete and rule-governed, and pricing requires no-arbitrage and the correct change of measure (we detail these stylized facts in §3). These tensions motivate the *structure-informed* designs—GBM forward processes [40], factor structure [14], arbitrage penalties [38], risk-neutral drift corrections—that recur throughout.

3 SYNTHETIC DATA AND TIME-SERIES GENERATION

The largest cluster targets data that passes for real. Two design philosophies recur. The first *reshapes* data so off-the-shelf diffusion applies: Takahashi and Mizuno [71] apply a wavelet transform mapping price, volume, and spread into image-like representations, train a DDPM, and invert it, recovering fat tails and clustering. The second *bakes in* financial structure: Kim et al. [40] replace Gaussian corruption with a geometric-Brownian-motion forward process, improving heavy-tail and clustering fidelity. A pattern recurs across such studies—which we state as a working hypothesis rather than a measured result, given the comparability problems of §10—that inductive bias (priors in the noise process or representation) tends to help more than architecture tuning alone. The fidelity bar is unusually demanding: a financial generator must reproduce a battery of *stylized facts* simultaneously—fat (power-law) tails, volatility clustering, the leverage effect, weak return autocorrelation alongside strong autocorrelation of absolute returns, and gain/loss asymmetry—none of which a low reconstruction error guarantees. The multivariate case additionally demands faithful cross-sectional dependence, motivating the factor- and correlation-structured designs that recur in §5–7.

Machinery differs by modality: low-frequency univariate and panel returns, the most-studied setting, are the home of stylized-fact evaluation [14, 71]; high-frequency microstructure data (§4) are discrete and event-driven, demanding event- or byte-level models [47, 87]; mixed-type tabular records (credit, fraud, holdings) require combined continuous/categorical diffusion [42, 61]; and structured objects such as implied-volatility surfaces or yield curves need no-arbitrage constraints built in [22, 38]. "Diffusion for finance" is thus not one method but a family, specialized per modality from the cross-domain backbones of §2.

High-dimensional generation. Portfolio and risk applications need realistic *dependence*. Chen et al. [14] embed a latent factor structure into diffusion, generating high-dimensional returns that respect a factor covariance, with theoretical guarantees and small-sample gains. Conditioning turns a generator into a scenario engine: CoFinDiff [72] controls trend and realized volatility with lower control error and more diversity than GAN/VAE baselines, and downstream hedging gains. Fidelity is ultimately judged by use: Fiaingen [60] reports train-on-synthetic/test-on-real near parity with real data, and comparative studies position diffusion against TimeGAN/VAE as a privacy-preserving substitute [33]; general style-guided TS diffusion [70] supplies further backbone machinery.

Tabular, fraud, privacy. Much financial data is mixed-type tabular. TabDDPM [42], latent-space TabSyn [84], and its rectified-flow analogue RecTable [21] define the general state of the art; FinDiff [61], presented at ICAIF, specializes it to financial tables (credit, expenditure, fund composition) with strong fidelity, utility, and privacy versus TVAE/CTGAN/TabDDPM. Two extensions address the field's defining difficulty—class imbalance: Imb-FinDiff [63] adds a joint denoising-plus-label objective for minority (fraud) synthesis, and EntTabDiff [50] conditions on *entities* to improve fidelity and reduce entity-level leakage. For fraud, EmDT [43] trains per-cluster diffusion transformers and latent flow-diffusion [36] targets minority oversampling. Privacy is moving from proxies to formal guarantees: DP-FinDiff [64] and federated DP-FedTabDiff [62] train under differential privacy, and Zuo et al. [88] benchmark the privacy–utility frontier. The non-diffusion frontier (adversarial synthetic data [80]) remains the comparison point.

Relation to classical and deep baselines. The comparison point is not only other deep generators but the calibrated stochastic models finance already trusts: GARCH-family volatility [11], stochastic-volatility [30] and rough-volatility [26] models for returns, and Hawkes/point-process models [7] for order flow. A diffusion generator earns its place only if it matches or beats a well-calibrated model from this class on the stylized facts that matter; most surveyed work benchmarks against deep baselines and reports this classical comparison only sparingly—a gap we flag. Among deep baselines, GANs—Quant GANs [79], Sig-Wasserstein GANs [54], signature-MMD models [53]—first reproduced stylized facts convincingly but train unstably, collapse modes, and condition awkwardly, while VAEs train stably but oversmooth the heavy tails finance cares about. Across the surveyed studies the *reported* pattern is consistent: diffusion matches or exceeds these baselines on stylized-fact and downstream metrics while conditioning more naturally, at the cost of slower sampling. We emphasize this is a synthesis of papers' self-reported comparisons, not an independently verified ranking (§10), so we read it as a hypothesis explaining the field's rapid migration rather than a settled result.

4 MARKET MICROSTRUCTURE AND LOB SIMULATION

Simulating the LOB and its order flow is harder than low-frequency generation: the data are high-frequency, discrete, bursty, and rule-governed, and a useful simulator must be *responsive*—reacting plausibly to an agent's own orders, so it can serve backtesting, market-impact studies, and agent-based experiments.

Order flow and representations. TRADES [10] generates order flow with a transformer diffusion conditioned on market state, reporting predictive-score gains; since the book reconstructs from the flow, realism propagates. "Painting the Market" [6] renders the LOB as an image and applies inpainting, generating long horizons in parallel and avoiding the error accumulation of autoregressive simulators, scoring well on LOB-Bench. ByteGen [47] forgoes tokenization, generating events directly in byte space with a hybrid Mamba–Transformer.

Volume, counterfactuals, and control. Beyond price levels, the volume profile governs liquidity and impact: DiffVolume [77] generates controllable volume snapshots, and DiffLOB [78] produces regime-conditioned LOB trajectories for microstructure what-if stress. DigMA [35] pairs a conditional diffusion of the market state

with a financially informed meta-agent for controllable, responsive order generation. Event-stream models (LOBDIF [87]) predict the joint timing and type of events with skip-step sampling for latency. Diffusion also serves upstream as a market-series *denoiser* [76], and lower-frequency dynamics simulation yields portfolio-ready covariances [45]. Interpretable evaluation uses optimal transport to test power-law reproduction [28].

The requirement that separates microstructure simulation from sequence generation is *responsiveness*: under a backtest or market-impact study the book must react plausibly to the agent's own orders, not replay history. Conditioning on the evolving market state—and, in DigMA [35], on an explicit agent—is the route, with counterfactual generation [77, 78] its controllable form. Latency (thousand-step sampling is infeasible at high frequency [87]) and the discrete, rule-governed nature of order events—which byte- and event-level models [47, 87] address more directly than image framings—temper progress.

This work sits in a longer arc—from hand-built agent-based and point-process models (e.g. Hawkes-driven order flow [7]), through GANs, to the diffusion frontier, which packages stable training, broad coverage, and flexible conditioning. Diffusion simulators are increasingly substrates for *agent-based* experiments [10, 35]: a calibrated, responsive generator lets researchers inject hypothetical agents or shocks and observe plausible reactions, which counterfactual and regime-conditioned designs [77, 78] enable. The open evaluation question—certifying that a simulator's *reaction* to an unseen agent is realistic, not merely that unconditional samples match history—remains the key obstacle to high-stakes strategy validation.

5 RISK, SCENARIO GENERATION, AND STRESS TESTING

Controllability pays off most in risk, where the goal is to populate *tails* and *what-ifs* absent from history. Guidance [20, 32] yields samples *given* a crash intensity or regime; the same heavy-tail fidelity that aids generation is what *would* make the resulting Value-at-Risk and Expected-Shortfall estimates credible. But the tails are exactly where generators most often fail their fidelity checks (§10), so this credibility must be validated, not assumed—a generator that thins tails silently understates risk. Evidence is strongest when stress scenarios are tested on chronological or held-out crisis periods and against classical volatility, extreme-value, or factor-stress baselines; many papers still report only deep generator comparisons. Methodologically, the generator is a Monte Carlo engine: VaR and ES/CVaR and their sensitivities are estimated from sampled scenarios, optionally conditioned on a stressed regime, so tail quality directly sets the capital number—which is why recent methods add tail-weighted objectives or regime conditioning [5, 75] rather than tuning for average-case realism. Scenario generation is most useful wired into a decision: diffusion-augmented RL [18] trains robust policies against generated stress, and MARCD [5] couples a regime-aligned diffusion generator (a hidden Markov model) to a CVaR-constrained allocator with a tail-weighted objective, reporting large drawdown reductions. Structure-aware generation applies stress coherently: factor-structured diffusion [14] shocks factors while preserving dependence; compressed-sensing-accelerated diffusion in a low-dimensional macro space [27] reproduces left-tail quantiles for backtesting; diffusion scenario trees [83] feed multi-stage stochastic optimization; and neural Lévy jump-diffusion [75] targets jump-dominated tails. Regulated stress testing could use richer, internally consistent scenario sets, but a scenario generator used for capital must itself be validated and auditable (Section 11)—in our view the main barrier between these results and regulated deployment.

6 PRICING, HEDGING, AND CALIBRATION

Pricing imposes the hardest economic constraints: no-arbitrage and the change of measure from the physical ($\mathbb{P}$) to the risk-neutral ($\mathbb{Q}$) world. A change of measure shifts the SDE drift [69], which in score/ϵ terms is an additive correction to the reverse dynamics; Tiwari [74] realize exactly this as a score-function shift enforcing the martingale condition, making the measure change explicit rather than approximate (cf. the built-in risk-neutral transition of Quant GANs [79]). Zhao and Shen [86] price exotic and path-dependent options (European, arithmetic Asian) with conditional diffusion, honestly reporting degraded accuracy on *extreme* scenarios—the crux for risk-sensitive products. Jin and Agarwal [38] generate arbitrage-aware implied-volatility surfaces one day ahead by penalizing arbitrage violations in training, beating GAN-based surface models. Pricing evidence is therefore most credible when it reports arbitrage-violation rates and tail errors, not only average pricing loss against neural or GAN baselines.

Fixed income and foundations. Diffusion has reached rates: Fukunishi et al. [22] generate arbitrage-aware yield-curve term structures with macroeconomic conditioning, and Zhang et al. [85] synthesize language-conditioned Treasury-futures series. More foundationally, Gao et al. [24] generate SDE solution paths *without* known drift and diffusion coefficients—the calibration setting when the model is unknown—and the HJB view [9] links diffusion to the optimal-investment and hedging equations. Calibration recurs in distributional form—arbitrage-consistent surfaces [38] and term structures [22], or learning unknown SDE coefficients when the model class is uncertain [24]. Deep hedging [12] is the downstream decision these generators serve: with a faithful $\mathbb{Q}$-generator, better training distributions yield better learned hedges (as CoFinDiff [72] shows) and Greeks follow by simulation on paths with frictions and jumps. Neural PDE solvers for (rough) volatility pricing [56] are a complementary, non-generative route. The unresolved frontier is enforcing *exact* no-arbitrage and the martingale property by construction rather than by penalty. Three routes trade off differently: soft penalties [38] are simple to add but only approximate and can still admit arbitrage; *projecting* samples onto the no-arbitrage set guarantees feasibility but distorts the learned distribution and its tails; and *constrained or guided sampling* imposes the constraint along the reverse SDE, at higher sampling cost. Achieving exactness while preserving tail accuracy, and at latencies real-time pricing can afford, remains open.

7 FORECASTING AND PORTFOLIO CONSTRUCTION

Diffusion models are *probabilistic* forecasters—they produce predictive distributions, which is what portfolio construction and risk budgeting consume. TimeGrad [58] is the standard baseline; in finance, Diffolio [17] adds hierarchical (asset- and market-level) attention and a correlation-guided regularizer, so the predictive distribution respects cross-sectional dependence, improving both accuracy and downstream Sharpe and certainty-equivalent. Allocation follows two routes. *Predict-then-optimize*: factor-conditioned diffusion learns the cross-sectional return distribution and optimizes mean-variance/mean-CVaR portfolios [23], with a bias–variance analysis of the conditioning-factor count [8]. *Direct* policy learning: model-free dynamic selection beats Markowitz [1], building on decision-as-generation [2, 37]; the flow-matching form of this idea, FlowHFT [48], learns an imitation-learning trading policy for high-frequency trading that adapts across market regimes. Relational and augmentation variants include DiffSTOCK [19] (conditioning on indicators and inter-stock relations) and DiffsFormer [25] (factor augmentation); chart-image forecasting [44] illustrates the role of representation. A third, indirect route is *augmentation*: synthetic paths enlarge thin training sets for downstream predictors and reinforcement-learning agents [25, 53], and the generate-then-optimize pipeline of §5 [5, 18] closes the loop by optimizing the allocation directly against generated scenarios. Across all three routes the strongest results couple a financial inductive bias to the generator; claims are weaker when they omit transaction costs, simple chronological rebalancing baselines, or regime-shifted test periods. A separate open question is how these diffusion forecasters compare against the large pretrained time-series foundation models now emerging as a competing paradigm. The recurring caveat is evaluation: portfolio metrics are especially vulnerable to look-ahead bias and single-backtest overfitting, a theme we turn to next.

8 EMERGING AND CROSS-DOMAIN APPLICATIONS

Beyond the five core tasks, adjacent applications are maturing along one recurring pattern: a technique proven on a structurally similar problem is specialized with a financial prior. *Fixed income* has its first generative treatments (arbitrage-aware term structures [22], Treasury-futures synthesis [85]); *macro-conditioned* generation links returns to macro state [23, 80]; and scenario trees [83] connect diffusion forecasts to multistage optimization.

9 CROSS-CUTTING THEMES

Read across tasks, four threads organize the field. **Conditioning is the norm, not the exception.** Recent finance-specific work—controllable generation [72], counterfactual microstructure [77, 78], stress scenarios [5, 18], and forecast-conditioned allocation [17, 23]—is conditional, because controllability is what finance wants from a generator. **Inductive bias appears to matter more than architecture tuning.** Where one design clearly outperforms another, the gain typically comes from encoding financial structure—GBM noising [40], factor covariance [14], arbitrage penalties [38], risk-neutral drift shifts [74], correlation regularizers [17]—into the noise process, the prior, or the loss; we offer this as a hypothesis the field's inconsistent evaluation (§10) cannot yet confirm. **Representation is a design lever.** Wavelet-to-image [71], LOB-as-image [6], byte-space [47], latent [59, 84], and chart-image [44] framings repeatedly unlock generic diffusion machinery for financial objects. **Generation increasingly serves a decision.** The frontier couples the generator to control—deep hedging [12, 72], CVaR allocation [5], multistage stochastic optimization [83], and the diffusion-as-policy line [2, 37]—rather than treating realistic samples as an end in themselves.

How to choose. Let the binding constraint, not the backbone, pick the method: stylized-fact batteries for synthetic returns, responsiveness tests for LOB simulators, Expected Shortfall under held-out crises for stress generators, arbitrage-violation rates for pricing, and chronological out-of-sample tests for portfolio models. When latency binds, prefer few-step samplers from the outset [66, 67].

Deployment considerations. Three constraints cut across all tasks: *latency* (few-step and skip-step sampling are load-bearing for real-time use [49, 66, 67, 87]); *data governance* (synthetic sensitive records require privacy and membership-risk checks [62, 64, 88]); and *validation* (generators feeding capital, valuation, or surveillance inherit model-risk-management duties).

10 EVALUATION, BENCHMARKS, REPRODUCIBILITY

Evaluation is the weakest link, and matters more in finance than in vision: a generator that looks good by a generic metric can misprice tail risk or leak look-ahead information. *Stylized-fact* fidelity (excess kurtosis, the Hill tail index, autocorrelation of returns vs. absolute returns) is the first-line test [40, 71], with optimal-transport distances a more interpretable, mechanism-attributing alternative for market simulators [28]; these checks are necessary but largely marginal. *Distributional/downstream* metrics probe joint and task-relevant structure: discriminative/predictive scores [10] and train-on-synthetic/test-on-real [60], the operational standard for replacing real data. Microstructure has LOB-Bench [6], though *responsiveness* (impact realism under the simulated agent's own actions) remains non-standardized. *Privacy* is first-class when synthetic data substitutes for sensitive records [61, 64, 88]. These form four metric families—marginal (stylized facts), joint/distributional (discriminative scores), task (downstream utility), and privacy—each guarding a different failure (Figure 3): a model can ace marginals yet fail joint dependence, or attain high utility yet leak membership. A credible study reports at least one from each; reporting only marginals, still common, is the easiest bar to game.

Three pitfalls undermine comparability. (i) *Heterogeneous data and metrics*: papers use different assets, periods, and scores, so reported superiority rarely transfers. (ii) *Leakage and look-ahead bias*: full-sample normalization or conditioning on information unavailable at decision time inflates results, especially in forecasting and portfolio studies. (iii) *In-distribution evaluation*: testing on the training regime hides the non-stationarity failure that matters in deployment. We recommend chronological splits, regime-shifted test sets, fixed public benchmarks, and released code; the absence of a shared, finance-specific benchmark suite is the single biggest obstacle to measurable progress.

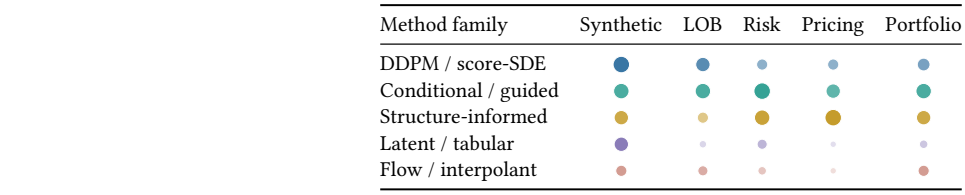


Figure 2: Coverage map of the surveyed literature. Each retained work was coded by primary method family and primary financial task; marker area and darkness increase with the number of coded works *in this survey* in each method × task cell (coarsely binned: small ≤ 2, medium 3–5, large > 5). It is not a bibliometric count of the wider literature.

Table 1: Representative finance works using learned generative diffusion, score-based, or flow-matching/rectified-flow models, with methodological backbone/representation and explicit conditioning ("C"). Classical "diffusion" SDE or PDE models without a learned generative diffusion/flow sampler are excluded from this table. The final column is the headline comparison *as reported by each cited work*—self-reported, protocol-heterogeneous, and not independently verified, hence indicative rather than a ranking (§10); entries are drawn from the same coding sheet summarized in Figure 2.

Work	Task	Modality	Backbone / representation	C	Reported comparison
Takahashi–Mizuno [71]	Synthetic gen	Returns	Wavelet→image DDPM		Fat tails, clustering
Diffusion Factor [14]	Hi-dim returns	Panel returns	Latent factor diffusion		Factor cov. + theory
CoFinDiff [72]	Controllable gen	Returns	Cond. (trend/vol)	✓	> GAN/VAE control err.
FinDiff [61]	Tabular	Mixed-type	Tabular diffusion	✓	> TVAE/CTGAN/TabDDPM
Imb-FinDiff [63]	Tabular (imbalance)	Mixed-type	Denoise + label obj.	✓	Minority (fraud) synth.
DP-FinDiff [64]	Private tabular	Mixed-type	Diffusion under DP	✓	Formal DP budget
TRADES [10]	LOB sim	Order flow	Transformer diffusion	✓	Predictive-score gain
Painting [6]	LOB sim	LOB image	Image inpainting	✓	LOB-Bench
DiffVolume [77]	LOB volume	Volume snapshot	Cond. counterfactual	✓	Counterfactual control
DiffLOB [78]	LOB counterfactual	LOB trajectory	Regime-conditioned	✓	Regime what-ifs
LOBDIF [87]	LOB event stream	Event stream	Event + skip-step		Joint timing/type
MARCD [5]	Stress + CVaR	Returns	Regime (HMM) + CVaR	✓	Drawdown reduction
Risk-neutral [74]	Pricing	Paths	Score-shift to $\mathbb{Q}$	✓	Martingale enforced
IV surface [38]	Pricing	Vol. surface	Arbitrage-penalized	✓	> GAN surfaces
TF-CoDiT [85]	Fixed income	Treasury futures	Latent diffusion transformer	✓	Low-data synthesis
Diffolio [17]	Forecast/portfolio	Panel returns	Hier. attn + corr. reg.	✓	Sharpe/CE gain
Factor-cond. [23]	Portfolio	Cross-section	Factor-conditioned	✓	Mean-var/CVaR
Dyn. portfolio [1]	Portfolio	Returns	Model-free policy		> Markowitz
FlowHFT [48]	Trading (flow)	HF features	Flow-matching policy	✓	Regime-adaptive

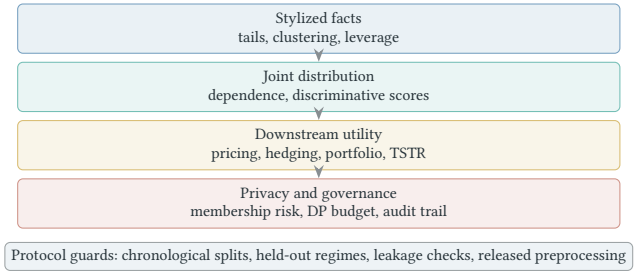


Figure 3: Evaluation stack for financial diffusion models. Each layer catches a different failure mode; credible studies should report at least one check at each layer rather than relying on marginal stylized facts alone.

A recommended reporting protocol. Concretely, we suggest each study report: (i) the exact data source, asset universe, and date range; (ii) a *chronological* train/test split with a held-out regime or crisis; (iii) at least one stylized-fact metric, discriminative/predictive score, and downstream (train-on-synthetic, test-on-real) metric, since each probes a different failure mode; (iv) for private data, a stated privacy notion and budget rather than an empirical proxy alone; and (v) released code and preprocessing. None of these is universal today, which is why headline claims of superiority so rarely reproduce across papers. To make Table 1 auditable and answer the code-availability question practitioners ask first, the companion artifact for release records paper metadata, method/task labels, modality, conditioning, evaluation protocol, publication status, and verifiable code/dataset availability.

11 OPEN PROBLEMS

Tail and extreme-event fidelity. Decision-relevant mass lives in the tails, yet strong results degrade there [86], and risk numbers inherit any tail thinning. Objectives and noise schedules that explicitly target tail accuracy, and rare-event conditioning, are open; structure-informed designs [14, 40] help but do not solve it. **Non-stationarity.** Out-of-regime generalization and guarantees under shift are largely unaddressed; existing guarantees are narrow [14]. **Hard economic constraints.** Enforcing exact no-arbitrage and the $\mathbb{Q}$-martingale property by construction (not penalty [38]) is a prerequisite for front-office adoption. **Efficiency.** Few-step sampling [49, 66, 67] for real-time/HFT is underexplored on heavy-tailed targets and is essential for responsive simulation. **Decision**

integration. Coupling generation to control [9, 18, 24] is promising but immature. **Governance.** Model risk, auditability, and formal privacy [64] gate regulated use. **Causality and counterfactuals.** Most models are correlational generators; counterfactual microstructure [77, 78] and controllable agents [35] gesture toward intervention, but principled causal diffusion—answering "what if this policy or shock occurred"—is largely unexplored and is precisely what stress testing requires. **Theory.** Approximation and sample-complexity guarantees remain narrow: factor-structure results [14] and the control-theoretic view [9] are starting points, but error bounds under non-stationarity and for heavy-tailed targets are open. **Benchmarks.** A shared finance-specific suite—returns, LOB [6], tabular [61], and risk—with chronological, regime-aware splits would do for this field what community benchmarks did for vision diffusion.

12 CONCLUSION

Diffusion models have earned a permanent place in the quantitative-finance toolkit, mapping their stable training, mode coverage, conditioning, and SDE structure onto finance's needs across data generation, microstructure simulation, risk, pricing, and portfolio decisions. The evidence is encouraging, but the field's progress is harder to measure than to produce: tail fidelity degrades where finance cares most, evaluation is fragmented and frequently in-distribution, economic constraints are penalized rather than enforced, sampling is too slow for real-time use, and governance for regulated deployment is barely begun. The next phase depends less on new architectures than on discipline: finance-specific benchmarks with chronological and regime-aware splits, tail- and arbitrage-aware objectives, formal privacy and model-risk treatment, and tighter coupling of generation to the decision it serves.

Scope and limitations. We emphasize breadth over exhaustive coverage of any one task; multimodal generation, graph-based systemic risk, and crypto-specific work receive only brief treatment. We expect the method × task map (Figures 1–2) to fill rapidly, especially in hard-constrained pricing and tail-faithful risk.

REFERENCES

[1] Ahmad Aghapour, Erhan Bayraktar, and Fengyi Yuan. 2025. Solving Dynamic Portfolio Selection Problems via Score-Based Diffusion Models. *arXiv preprint arXiv:2507.09916* (2025).

[2] Anurag Ajay, Yilun Du, Abhi Gupta, Joshua Tenenbaum, Tommi Jaakkola, and Pulkit Agrawal. 2023. Is Conditional Generative Modeling all you need for Decision-Making?. In *ICLR*. arXiv:2211.15657.

[3] Michael S. Albergo, Nicholas M. Boffi, and Eric Vanden-Eijnden. 2023. Stochastic Interpolants: A Unifying Framework for Flows and Diffusions. *arXiv preprint arXiv:2303.08797* (2023).

[4] Juan Miguel Lopez Alcaraz and Nils Strodthoff. 2022. Diffusion-based Time Series Imputation and Forecasting with Structured State Space Models. *Transactions on Machine Learning Research* (2022). arXiv:2208.09399.

[5] Ali Atiah Alzahrani. 2025. Multi-Agent Regime-Conditioned Diffusion (MARCD) for CVaR-Constrained Portfolio Decisions. *arXiv preprint arXiv:2510.10807* (2025).

[6] Alfred Backhouse, Kang Li, Jakob Foerster, Anisoara Calinescu, and Stefan Zohren. 2025. Painting the Market: Generative Diffusion Models for Financial Limit Order Book Simulation and Forecasting. *arXiv preprint arXiv:2509.05107* (2025).

[7] Emmanuel Bacry, Iacopo Mastromatteo, and Jean-François Muzy. 2015. Hawkes Processes in Finance. *Market Microstructure and Liquidity* 1, 1 (2015), 1550005.

[8] Avi Bagchi, Michael Tesfaye, and Om Shastri. 2026. Factor Dimensionality and the Bias-Variance Tradeoff in Diffusion Portfolio Models. *arXiv preprint arXiv:2603.10385* (2026).

[9] Julius Berner, Lorenz Richter, and Karen Ullrich. 2024. An Optimal Control Perspective on Diffusion-Based Generative Modeling. *Transactions on Machine Learning Research* (2024). arXiv:2211.01364.

[10] Leonardo Berti, Bardh Prenkaj, and Paola Velardi. 2025. TRADES: Generating Realistic Market Simulations with Diffusion Models. *arXiv preprint arXiv:2502.07071* (2025).

[11] Tim Bollerslev. 1986. Generalized Autoregressive Conditional Heteroskedasticity. *Journal of Econometrics* 31, 3 (1986), 307–327.

[12] Hans Bühler, Lukas Gonon, Josef Teichmann, and Ben Wood. 2019. Deep Hedging. *Quantitative Finance* 19, 8 (2019). arXiv:1802.03042.

[13] Raju Challagundla, Mohsen Dorodchi, Pu Wang, and Minwoo Lee. 2025. Synthetic Tabular Data Generation: A Comparative Survey for Modern Techniques. *arXiv preprint arXiv:2507.11590* (2025).

[14] Minshuo Chen, Renyuan Xu, Yumin Xu, and Ruixun Zhang. 2025. Diffusion Factor Models: Generating High-Dimensional Returns with Factor Structure. *arXiv preprint arXiv:2504.06566* (2025).

[15] Yu Chen, Marin Biloš, Sarthak Mittal, Wei Deng, Kashif Rasul, and Anderson Schneider. 2024. Recurrent Interpolants for Probabilistic Time Series Prediction. *arXiv preprint arXiv:2409.11684* (2024).

[16] Yifan Chen, Mark Goldstein, Mengjian Hua, Michael S. Albergo, Nicholas M. Boffi, and Eric Vanden-Eijnden. 2024. Probabilistic Forecasting with Stochastic Interpolants and Föllmer Processes. In *ICML*. arXiv:2403.13724.

[17] So-Yoon Cho, Jin-Young Kim, Kayoung Ban, Hyeng Keun Koo, and Hyun-Gyoon Kim. 2025. Diffolio: A Diffusion Model for Multivariate Probabilistic Financial Time-Series Forecasting and Portfolio Construction. *arXiv preprint arXiv:2511.07014* (2025).

[18] Himanshu Choudhary, Arishi Orra, and Manoj Thakur. 2025. Diffusion-Augmented Reinforcement Learning for Robust Portfolio Optimization under Stress Scenarios. *arXiv preprint arXiv:2510.07099* (2025).

[19] Divyanshu Daiya, Monika Yadav, and Harshit Singh Rao. 2024. DiffSTOCK: Probabilistic Relational Stock Market Predictions using Diffusion Models. In *ICASSP*. arXiv:2403.14063.

[20] Prafulla Dhariwal and Alexander Quinn Nichol. 2021. Diffusion Models Beat GANs on Image Synthesis. In *NeurIPS*. arXiv:2105.05233.

[21] Masane Fuchi and Tomohiro Takagi. 2025. RecTable: Fast Modeling Tabular Data with Rectified Flow. *arXiv preprint arXiv:2503.20731* (2025).

[22] Yosuke Fukunishi, Haorong Qiu, Akihiko Takahashi, and Fan Ye. 2025. Generating the Term Structure of Interest Rates with Diffusion Models. *Expert Systems with Applications (forthcoming); CIRJE Discussion Paper CF-1262* (2025).

[23] Xuefeng Gao, Mengying He, Xuedong He, and Jiale Zha. 2025. Factor-Based Conditional Diffusion Model for Contextual Portfolio Optimization. *arXiv preprint arXiv:2509.22088* (2025).

[24] Xuefeng Gao, Jiale Zha, and Xun Yu Zhou. 2025. Generating Solution Paths of Markovian Stochastic Differential Equations using Diffusion Models. *arXiv preprint arXiv:2509.08731* (2025).

[25] Yuan Gao, Haokun Chen, Xiang Wang, Zhicai Wang, Xue Wang, Jinyang Gao, and Bolin Ding. 2024. DiffsFormer: A Diffusion Transformer on Stock Factor Augmentation. *arXiv preprint arXiv:2402.06656* (2024).

[26] Jim Gatheral, Thibault Jaisson, and Mathieu Rosenbaum. 2018. Volatility is Rough. *Quantitative Finance* 18, 6 (2018), 933–949.

[27] Zhengyi Guo, Jiatu Li, Wenpin Tang, and David D. Yao. 2025. Diffusion Generative Models Meet Compressed Sensing, with Applications to Imaging and Finance. *arXiv preprint arXiv:2509.03898* (2025).

[28] Ryuji Hashimoto and Kiyoshi Izumi. 2025. Towards Realistic and Interpretable Market Simulations: Factorizing Financial Power Law using Optimal Transport. *arXiv preprint arXiv:2507.09863* (2025).

[29] Panjing He, Mingyue Cheng, Li Li, and XiaoHan Zhang. 2025. TimeFlow: Towards Stochastic-Aware and Efficient Time Series Generation via Flow Matching Modeling. *arXiv preprint arXiv:2511.07968* (2025).

[30] Steven L. Heston. 1993. A Closed-Form Solution for Options with Stochastic Volatility with Applications to Bond and Currency Options. *The Review of Financial Studies* 6, 2 (1993), 327–343.

[31] Jonathan Ho, Ajay Jain, and Pieter Abbeel. 2020. Denoising Diffusion Probabilistic Models. In *NeurIPS*. arXiv:2006.11239.

[32] Jonathan Ho and Tim Salimans. 2022. Classifier-Free Diffusion Guidance. In *NeurIPS Workshop on Deep Generative Models*. arXiv:2207.12598.

[33] Christophe D. Hounwanou and Yae Ulrich Gaba. 2025. Deep Generative Models for Synthetic Financial Data: Applications to Portfolio and Risk Modeling. *arXiv preprint arXiv:2512.21798* (2025).

[34] Yang Hu, Xiao Wang, Zezhen Ding, Lirong Wu, Huatian Zhang, Stan Z. Li, Sheng Wang, Jiheng Zhang, Ziyun Li, and Tianlong Chen. 2024. FlowTS: Time Series Generation via Rectified Flow. *arXiv preprint arXiv:2411.07506* (2024).

[35] Yu-Hao Huang, Chang Xu, Yang Liu, Weiqing Liu, Wu-Jun Li, and Jiang Bian. 2026. Controllable Financial Market Generation with Diffusion Guided Meta Agent. In *AAAI*. arXiv:2408.12991, oral.

[36] Md. Tawfique Ihsan, Md. Rakibul Hasan Rafi, Ahmed Shoyeb Raihan, Imtiaz Ahmed, and Abdullahil Azeem. 2025. Boosting Predictive Performance on Tabular Data through Data Augmentation with Latent-Space Flow-Based Diffusion. *arXiv preprint arXiv:2511.16571* (2025).

[37] Michael Janner, Yilun Du, Joshua B. Tenenbaum, and Sergey Levine. 2022. Planning with Diffusion for Flexible Behavior Synthesis. In *ICML*. arXiv:2205.09991.

[38] Chen Jin and Ankush Agarwal. 2025. Forecasting Implied Volatility Surface with Generative Diffusion Models. *arXiv preprint arXiv:2511.07571* (2025).

[39] Tero Karras, Miika Aittala, Timo Aila, and Samuli Laine. 2022. Elucidating the Design Space of Diffusion-Based Generative Models. In *NeurIPS*. arXiv:2206.00364.

[40] Gihun Kim, Sun-Yong Choi, and Yeoneung Kim. 2025. A Diffusion-Based Generative Model for Financial Time Series via Geometric Brownian Motion. *arXiv preprint arXiv:2507.19003* (2025).

[41] Marcel Kollovieh, Abdul Fatir Ansari, Michael Bohlke-Schneider, Jasper Zschiegner, Hao Wang, and Yuyang Wang. 2023. Predict, Refine, Synthesize: Self-Guiding Diffusion Models for Probabilistic Time Series Forecasting. In *NeurIPS*. arXiv:2307.11494.

[42] Akim Kotelnikov, Dmitry Baranchuk, Ivan Rubachev, and Artem Babenko. 2023. TabDDPM: Modelling Tabular Data with Diffusion Models. In *ICML*. arXiv:2209.15421.

[43] En-Ya Kuo and Sebastien Motsch. 2026. EmDT: Embedding Diffusion Transformer for Tabular Data Generation in Fraud Detection. *arXiv preprint arXiv:2603.13566* (2026).

[44] Taegyeong Lee, Jiwon Park, Kyunga Bang, Seunghyun Hwang, and Ung-Jin Jang. 2025. Exploring Diffusion Models for Generative Forecasting of Financial Charts. *arXiv preprint arXiv:2509.02308* (2025).

[45] Andrew Lesniewski and Giulio Trigila. 2024. Beyond Monte Carlo: Harnessing Diffusion Models to Simulate Financial Market Dynamics. *arXiv preprint arXiv:2412.00036* (2024).

[46] Dawei Li, Yue Huang, Ming Li, Tianyi Zhou, Xiangliang Zhang, and Huan Liu. 2025. Generative Models for Synthetic Data: Transforming Data Mining in the GenAI Era. *arXiv preprint arXiv:2508.19570* (2025).

[47] Yang Li and Zhi Chen. 2025. ByteGen: A Tokenizer-Free Generative Model for Orderbook Events in Byte Space. *arXiv preprint arXiv:2508.02247* (2025).

[48] Yang Li, Zhi Chen, and Steve Yang. 2025. FlowHFT: Imitation Learning via Flow Matching Policy for Optimal High-Frequency Trading under Diverse Market Conditions. *arXiv preprint arXiv:2505.05784* (2025).

[49] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. 2023. Flow Matching for Generative Modeling. In *ICLR*. arXiv:2210.02747.

[50] Changshuo Liu and Canyao Liu. 2024. Entity-based Financial Tabular Data Synthesis with Diffusion Models. In *Proc. 5th ACM Int. Conf. on AI in Finance (ICAIF)*. doi:10.1145/3677052.3698625.

[51] Jing Liu, Zhenchao Ma, Zepu Wang, Chenxuanyin Zou, Jiayang Ren, Zehua Wang, Liang Song, Bo Hu, Yang Liu, and Victor C.M. Leung. 2025. A Survey on Diffusion Models for Anomaly Detection. *arXiv preprint arXiv:2501.11430* (2025).

[52] Xingchao Liu, Chengyue Gong, and Qiang Liu. 2023. Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow. In *ICLR*. arXiv:2209.03003.

[53] Chung I Lu and Julian Sester. 2024. Generative Modelling of Financial Time Series with Structured Noise and MMD-based Signature Learning. *arXiv preprint arXiv:2407.19848* (2024).

[54] Hao Ni, Lukasz Szpruch, Marc Sabate-Vidales, Baoren Xiao, Magnus Wiese, and Shujian Liao. 2021. Sig-Wasserstein GANs for Time Series Generation. In *Proc. 2nd ACM Int. Conf. on AI in Finance (ICAIF)*. arXiv:2111.01207.

[55] Alexander Quinn Nichol and Prafulla Dhariwal. 2021. Improved Denoising Diffusion Probabilistic Models. In *ICML*. arXiv:2102.09672.

[56] Antonis Papapantoleon and Jasper Rou. 2024. A Time-Stepping Deep Gradient Flow Method for Option Pricing in (Rough) Diffusion Models. *arXiv preprint arXiv:2403.00746* (2024).

[57] William Peebles and Saining Xie. 2023. Scalable Diffusion Models with Transformers. In *ICCV*. arXiv:2212.09748.

[58] Kashif Rasul, Calvin Seward, Ingmar Schuster, and Roland Vollgraf. 2021. Autoregressive Denoising Diffusion Models for Multivariate Probabilistic Time Series Forecasting. In *ICML*. arXiv:2101.12072.

[59] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2022. High-Resolution Image Synthesis with Latent Diffusion Models. In *CVPR*. arXiv:2112.10752.

[60] Jože M. Rožanec, Tina Žezlin, Laurentiu Vasiliu, Dunja Mladenić, Radu Prodan, and Dumitru Roman. 2025. Fiaingen: A Financial Time Series Generative Method Matching Real-World Data Quality. *arXiv preprint arXiv:2510.01169* (2025).

[61] Timur Sattarov, Marco Schreyer, and Damian Borth. 2023. FinDiff: Diffusion Models for Financial Tabular Data Generation. In *Proc. 4th ACM Int. Conf. on AI in Finance (ICAIF)*. arXiv:2309.01472.

[62] Timur Sattarov, Marco Schreyer, and Damian Borth. 2024. Federated Diffusion Modeling with Differential Privacy for Tabular Data Synthesis. *arXiv preprint arXiv:2412.16083* (2024).

[63] Timur Sattarov, Marco Schreyer, and Damian Borth. 2024. Imb-FinDiff: Conditional Diffusion Models for Class Imbalance Synthesis of Financial Tabular Data. In *Proc. 5th ACM Int. Conf. on AI in Finance (ICAIF)*. doi:10.1145/3677052.3698659.

[64] Timur Sattarov, Marco Schreyer, and Damian Borth. 2025. Privacy Preserving Diffusion Models for Mixed-Type Tabular Data Generation. *arXiv preprint arXiv:2512.00638* (2025).

[65] Jascha Sohl-Dickstein, Eric A. Weiss, Niru Maheswaranathan, and Surya Ganguli. 2015. Deep Unsupervised Learning using Nonequilibrium Thermodynamics. In *ICML*. arXiv:1503.03585.

[66] Jiaming Song, Chenlin Meng, and Stefano Ermon. 2021. Denoising Diffusion Implicit Models. In *ICLR*. arXiv:2010.02502.

[67] Yang Song, Prafulla Dhariwal, Mark Chen, and Ilya Sutskever. 2023. Consistency Models. In *ICML*. arXiv:2303.01469.

[68] Yang Song and Stefano Ermon. 2019. Generative Modeling by Estimating Gradients of the Data Distribution. In *NeurIPS*. arXiv:1907.05600.

[69] Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. 2021. Score-Based Generative Modeling through Stochastic Differential Equations. In *ICLR*. arXiv:2011.13456.

[70] Mingchun Sun, Rongqiang Zhao, Hengrui Hu, Songyu Ding, and Jie Liu. 2025. DS-Diffusion: Data Style-Guided Diffusion Model for Time-Series Generation. *arXiv preprint arXiv:2509.18584* (2025).

[71] Tomonori Takahashi and Takayuki Mizuno. 2025. Generation of Synthetic Financial Time Series by Diffusion Models. *Quantitative Finance* (2025). arXiv:2410.18897.

[72] Yuki Tanaka, Ryuji Hashimoto, Takehiro Takayanagi, Zhe Piao, Yuri Murayama, and Kiyoshi Izumi. 2025. CoFinDiff: Controllable Financial Diffusion Model for Time Series Generation. *arXiv preprint arXiv:2503.04164* (2025).

[73] Yusuke Tashiro, Jiaming Song, Yang Song, and Stefano Ermon. 2021. CSDI: Conditional Score-based Diffusion Models for Probabilistic Time Series Imputation. In *NeurIPS*. arXiv:2107.03502.

[74] Nilay Tiwari. 2026. Generative Diffusion Model for Risk-Neutral Derivative Pricing. *arXiv preprint arXiv:2603.20582* (2026).

[75] Ziyao Wang and Svetlozar T. Rachev. 2025. Neural Lévy SDE for State-Dependent Risk and Density Forecasting. *arXiv preprint arXiv:2509.01041* (2025).

[76] Zhuohan Wang and Carmine Ventre. 2024. A Financial Time Series Denoiser Based on Diffusion Model. *arXiv preprint arXiv:2409.02138* (2024).

[77] Zhuohan Wang and Carmine Ventre. 2025. DiffVolume: Diffusion Models for Volume Generation in Limit Order Books. *arXiv preprint arXiv:2508.08698* (2025).

[78] Zhuohan Wang and Carmine Ventre. 2026. DiffLOB: Diffusion Models for Counterfactual Generation in Limit Order Books. *arXiv preprint arXiv:2602.03776* (2026).

[79] Magnus Wiese, Robert Knobloch, Ralf Korn, and Peter Kretschmer. 2020. Quant GANs: Deep Generation of Financial Time Series. *Quantitative Finance* 20, 9 (2020). arXiv:1907.06673.

[80] Haochong Xia, Simin Li, Ruixiao Xu, Zhixia Zhang, Hongxiang Wang, Zhiqian Liu, Teng Yao Long, Molei Qin, Chuqiao Zong, and Bo An. 2026. Bayesian Robust Financial Trading with Adversarial Synthetic Market Data. *arXiv preprint arXiv:2601.17008* (2026).

[81] Yiyuan Yang, Ming Jin, Haomin Wen, Chaoli Zhang, Yuxuan Liang, Lintao Ma, Yi Wang, Chenghao Liu, Bin Yang, Zenglin Xu, Shirui Pan, and Qingsong Wen. 2024. A Survey on Diffusion Models for Time Series and Spatio-Temporal Data. *Comput. Surveys* (2024). arXiv:2404.18886.

[82] Xinyu Yuan and Yan Qiao. 2024. Diffusion-TS: Interpretable Diffusion for General Time Series Generation. In *ICLR*. arXiv:2403.01742.

[83] Stelios Zarifis, Ioannis Kordonis, and Petros Maragos. 2025. Diffusion-Based Scenario Tree Generation for Multivariate Time Series Prediction and Multistage Stochastic Optimization. *arXiv preprint arXiv:2509.14832* (2025).

[84] Hengrui Zhang, Jiani Zhang, Balasubramaniam Srinivasan, Zhengyuan Shen, Xiao Qin, Christos Faloutsos, Huzefa Rangwala, and George Karypis. 2024. Mixed-Type Tabular Data Synthesis with Score-based Diffusion in Latent Space. In *ICLR*. arXiv:2310.09656.

[85] Yingxiao Zhang, Jiaxin Duan, Junfu Zhang, and Ke Feng. 2026. TF-CoDiT: Conditional Time Series Synthesis with Diffusion Transformers for Treasury Futures. *arXiv preprint arXiv:2601.11880* (2026).

[86] Helin Zhao and Junchi Shen. 2025. Valuation of Exotic Options and Counterparty Games Based on Conditional Diffusion. *arXiv preprint arXiv:2509.13374* (2025).

[87] Zetao Zheng, Guoan Li, Deqiang Ouyang, Decui Liang, and Jie Shao. 2024. Limit Order Book Event Stream Prediction with Diffusion Model. *Data Science and Engineering* (2024). arXiv:2412.09631.

[88] Michael Zuo, Inwon Kang, Stacy Patterson, and Oshani Seneviratne. 2026. Measuring Privacy Risks and Tradeoffs in Financial Synthetic Data Generation. *arXiv preprint arXiv:2602.09288* (2026).